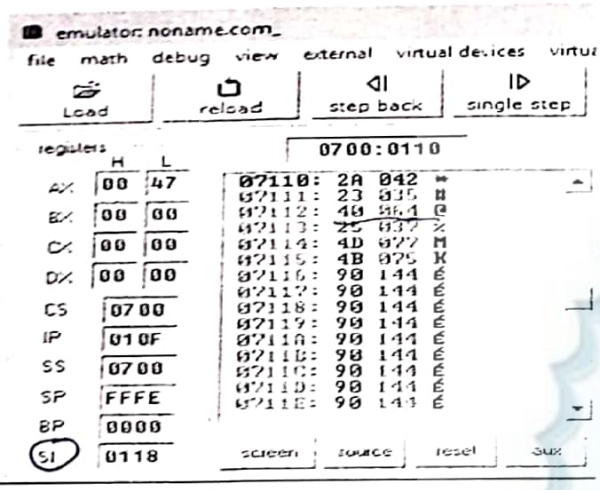
**Q1) Depending on the following snapshot of memory, answer the following questions (6pts)**




a) What is the physical address for the memory location pointed by SI?

07118 H

b) What is the ASCII code for symbol '@' as it shown in figure?

40

c) What is the DS value?

0700 H

5

d) Write the suitable instructions to swap between memory locations 07110H and 07111H?

```
mov ax, 700h
mov Ds, ax
mov si, 110h
mov cx, [si]
inc si
mov bx, [si]
EXCHG cx, bx
```

**Q2) Trace the following (10 pts)**

a-
```
org 100h
mov di,00
mov ax,54D
mov bl,2
L3: cmp al,10
jb L
div bl
add ah,30h
mov x[di],ah
inc di
mov ah,0
jmp L3
L: add al,30h
mov x[di],al
hlt

x db 3 dup(?)
```

di = 00

ah=30h

36h

X[3]=36

al= 36h ,ah= 00h

x= 30h, 31h, 31h, 36h

b-
```
mov al,1
mov ch,0
mov dx,20h
mov cl,2
L: add dl,cl
mul dl
loop L
```

Dx= ____ h

Cl= 0

Ax= ____

7

c-
```
MOV DL, 0
MOV CX, 3
MOV AL, 5
L3: SHR CL, 1
JNC L1
ADD DL, AL
L1: SHL AL, 1
CMP CX,0
JE L2
JMP L3
L2:
```

AL = 14h

20D

DL= 0FH

d-
```
MOV AX, 8000H
MOV BX, 0001H
SUB AX, BX
```

Ax= F

Cf= 0

Sf= 0

Pf= 1

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Student Name:

**Q3) Write a program to find how many numbers exist in the following array 'A1' that are less than or equal 6 then store the result in Y (8 pts)** A1 DB 3, 7, 8, 5, 21, 6

```
org 100h
mov cx= 6
mov si, offset A1
mov bL, 0
L2: mov Ah, [si]
    Cmp Ah, 6
    JBE L1:
    inc si
    JMP L2

L1: inc bL
    mov Y, bL
    hlt

A1 DB 3,7,8,5,21,6
Y DB ?
```

inc si
JMP L2

**Q4) Write an equivalent Assembly code for the following equation:**

**es = ((X+10) * Y)/Z (6 pts.)**

**lote: X, Y and Z are byte sizes (USING INDIRECT ADDRESSING MODE TO ACCESS THE ATA)**

```
org 100h
mov si, 0
mov al, X[si]
inc si
mov bL, X[si]
incsi
mov cL, X[si]
ADD aL, 10
mul bL ⇒ AX
Div AX, cL → of AX
mov res, AX
```

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Faculty of engineering technology
Computer Engineering department

*final assembly lab Exam(A)*

2nd Semester 2017/2018

Student No: [redacted]
Student Name: [redacted]

instructor name: Manal al_rawashdeh
Mohammad al_otari

Exam time:50 min

➢ Answer the following Squestion

**Question # 1:** Write an assembly code to input number (from 0 to 9) from keyboard then print "even" if number is even ,other wise print "add" Write complete program?( 10 pts)

**Question #2:** Trace the following (10 pts):

#1:

```
Org 100h
Mov ah,9
Lea dx,str
Int 21 h
Hlt
Str db " assembly",09,41h,42h,24h
```

[redacted]

#2:

```
org 100
Mov ah,9
Lea dx,x1
int 21H
hlt
X1 db "ab"
X2 db "cd5"
X3 db "ef5"
```

[redacted]

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#1

Mov ax ,3232h

Mov bx,2734h

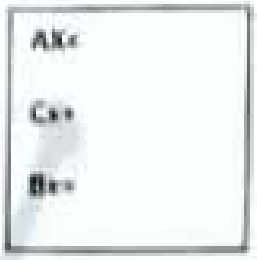
Push ax

Push bx

Pop cx

Pop bx

xchg




**Question #3:**

Choose the correct answer for following points: (14pts)

1. Assuming AL = 4 and BL = 5, the code:
   MOV AL, BL MOV BL, AL is equivalent to:
   a) MOV BL, AL
      MOV AL, BL
   c) MOV BL, 4
   d) MOV BL, 5
   e) XCHG AL, BL
2. Assuming AL = 1 and BL = -1, the instruction: MUL BL will produce:
   a) AX = 1 b) AX = -1 c) AX = 00FFh d) AX = FF00h e) AX = 0100h

3. Assuming AL = -1 and BL = -1, the instruction: DIV BL will produce:
   a) AX = 1 b) AX = -1 c) AX = 00FFh d) AX = FF00h e) AX = 0100h

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Faculty of engineering technology
Computer Engineering department

*final assembly lab Exam(A)*

Student No:
Student Name: [redacted]

2nd Semester 2017/2018
instruction name: Manal al [illegible]
Mohammed al [illegible]

Exam time:50 min

4. neg is equivalent to :
a)multi -1 b) Subtraction c) div d) add e)a+b

5. LEA is means
a) Load effective address b) Load address. c) Load physical address. d) None of them .

6. Push is means
a) Operand onto the top of the hardware supported register
b) places its operand onto the top of the hardware supported stack in memory
c) Operand onto the top of the hardware supported stack in memory
d) None above

7.test instruction is equivalent to:
a)cmp b)jnz c)shl d)shr e)all of them

**Question #4:**

**Write an assembly code to exchange the values in the two arrays using stack (push and pop instructions)? (10)**

**Question #5:**

**Suppose that DS=0200H, BX=0300H, and DI=400H. Determine the memory address accessed by each of the following instructions, assuming real mode operation(6pts)**

**(a) MOV AL, [1234H]**

**(b) MOV AX, [BX]**

**(c) MOV [DI], AL**

**The End**

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Computer Engineering Department
Assembly Lab Midterm Exam

Date: Student Name:

---

Question: Write a program that finds the count of negative even numbers in an array of 10 elements, and stores it in memory location **count**.

Note: an example is:

ARR[] = {2, 9, -1, 0, -5, 3, 8, -4, -2, 7}

count = 4

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Tuesday 11:00 - 14:00

هناء جمال ناهي اسماعيل

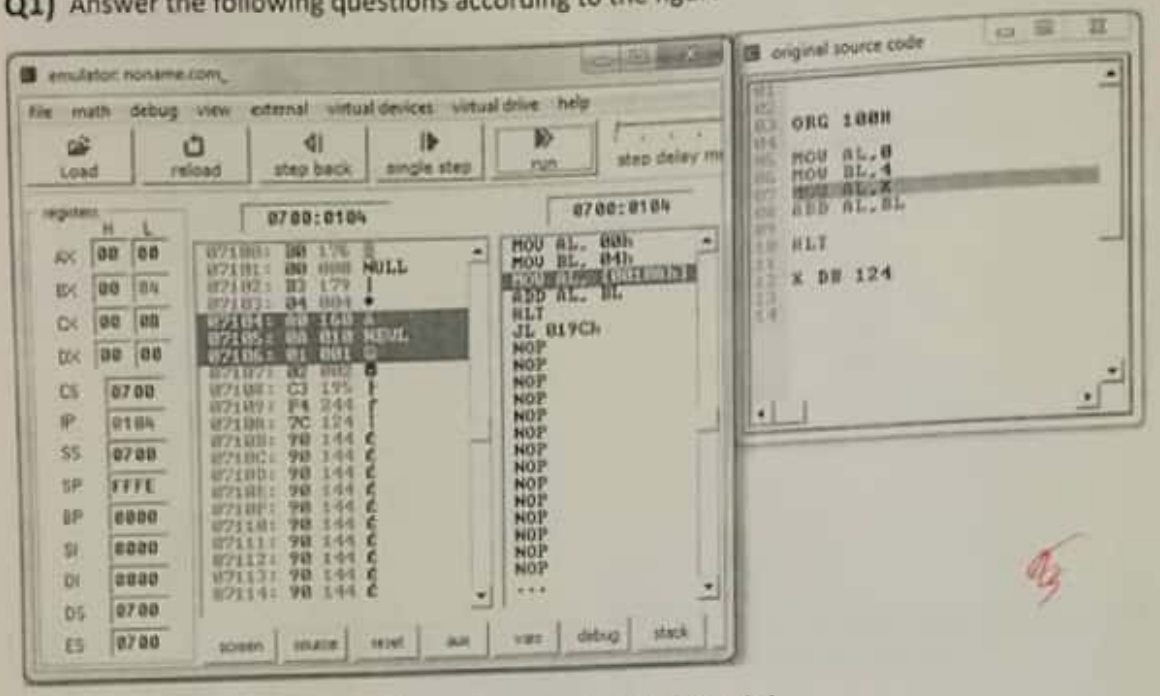
Al-Balqa Applied University
Computer Engineering Department
Student Name:

Microprocessor Systems Lab
Mid Exam (30%)

Faculty of Engineering Technology
Instructor: Eng. Nawal Alzabin

28/30

**Q1)** Answer the following questions according to the figure below




A) What is the physical address of the variable X? (1 Marks)

07104 H

B) What is the value of IP register after executing the highlighted instruction? (1 Marks)

107 H

C) If the machine code of the HLT instruction is F4H, what is the logical address of the HLT instruction? (1 Marks)

109 H

D) Fill the value of AL register and Flags after executing the instruction "ADD AL, BL". (2 Marks)

AL = 80 H

CF = 0, PF = 0, AF = 1, ZF = 0, SF = 1, OF = 1

**Q2)** *(5 Marks)* Read the word at address DS:600H, shift its value right and store it back to memory.

```
Org 100H
LEA SI, DS:600H
Mov AX, [SI]
SHR AX, 1
Mov [SI], AX
HLT
```



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Student Name: Tuesday 11:00 -14:00 ضياء جمال ناجي اسماعيل

**Q3) (6 Marks) Write an assembly code to exchange every BYTE which in an odd position in array x with the previous BYTE in the same array.**

| | SI | DI | | | | | |
|---|---|---|---|---|---|---|---|
| X | byte0 | byte 1 | byte2 | byte3 | byte4 | byte5 | Byte6 |

**Q4) Write an assembly code to read character from keyboard then do the following:**

1- If the character is 'a' then print your name on screen as follow:

ahmad ahmad
ahmad ahmad
ahmad ahmad
ahmad ahmad
ahmad ahmad

2- If the character is 'b' store it in variable W.

**Q5)** *(5 Marks)* Correct the following code that do the following: for each byte stored in memory from address 900H:100H to address 900H:110H, if the byte is multiple of 3, replace it with zero.

```
ORG 100H
MOV DS, 9000H      → Mov DX, 900 H
                     Mov DS, DX
MOV SI, 100H
MOV CX, 10
MOV bl,3
MOV AL, [SI]
L1: MUL BL         → L1: Div BL
CMP AL,0           → Cmp AH, 0
JNE L2             → JE L2
MOV [SI],0
L2:                → Inc SI
LOOP L1
HLT
```



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Q.3)
```
Org 100H
LEA SI, X
LEA DI, X+1
Mov CX, 4
L1:
Mov AL, [SI]
Mov BL, [DI]
Mov [DI], AL
Mov [SI], BL
Add SI, 2
Add DI, 2
Loop L1
Hlt
X DB 1,2,3,4,5,6,7,8
```

Q.4)
```
Org 100H
Mov CX, 5
Mov AH, 1
Int 21H
Cmp AL, 61H
JE L1
Cmp AL, 62H
JNE L2
Mov W, AL
Jmp L2
L1:
Mov AH, 2
Mov DX, 08H
Int 21H
Mov AH, 9
Mov DX, offset X
Int 21H
Mov AH, 2
Mov DL, 09H
Int 21H
Mov AH, 9
Mov DX, offset X
Int 21H
Mov AH, 2
Mov DL, 0AH
Int 21H
Mov DL, 0DH
Int 21H
Loop L1
L2:
HLT
W DB ?
X DB 'ahmad$'
```

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Student Name:

**Q3) Trace the following (10 pts)**

```
x dw 4020h,2284h,0aaffh
y db 7,9,5,1
mov bx, offset x
push [bx+1]
mov ax,[bx]
add bx,2
mov cx,[bx]
mov si, offset y
sub si,1
push [si]
pop bx
pop dx
```

ax= 4020h cx= 2284h
dx= AAFFh

```
mov   dl, 0
mov   cx, 3
mov   al, 5
l3:   shr   cl, 1
jnc   l1
add   dl, al
l1:   shl   al, 1
jcxz  l2
jmp   l3
l2:
```

al = 5 dl = K

```
array dw  2, 4, 6, 8
count dw  5
mov ax, 0
stc       ;set cf=1
mov cx, count
mov si, offset array
label1: adc ax, [si]
add si, 2
loop label1
label2:
```

ax = 1A cx = 0

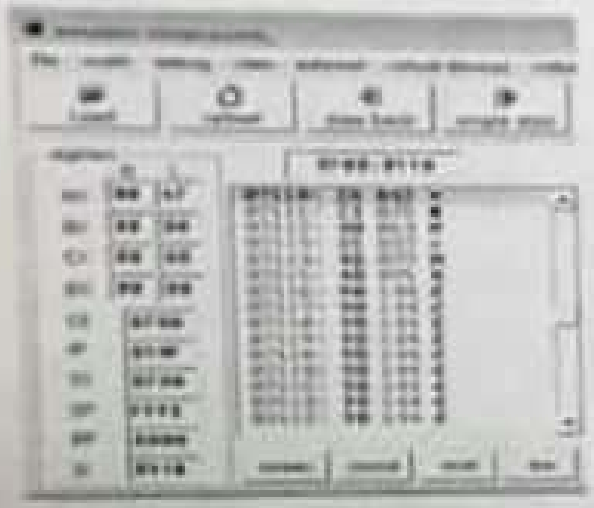
**Q4) Write an assembly program that counts number of (m) letter in the following string (8 pts)**

```
Str DB 'assembly programming Lang.'
```

```
org 100h
LEA SI, Str
mov cx, 26
mov AL, 0
L1: cmp [SI], 'm'
JNZ L2
Inc AL
L2: Loop L1
HLT
Str DB 'assembly programming Lang.'
```

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**Q1) Depending on the following snapshot of memory, answer the following questions (6pts)**



a) What is the physical address for the instruction pointed by IP?

27110h

b) What is the ASCII code for character 'M' as it shown in figure?

077h

c) What is the DS value?

0700 h

**Q2) write an assembly code to print *hello* on screen as shown below (6pts)**




```
org 100h
mov AH,09H
LEA DX, msg1
int 21h
mov AH, 02
mov DX, 0a10h
mov BH, 0
int 10h
mov ah, 09
LEA DX, msg2
int 21h
mov ah, 02
mov DX, 0B00h
mov BH, 0
int 10h
mov AH,
LEA DX, msg3
int 21h
H2h
```

```
msg1 DB "HE$"
msg2 DB "O$"
msg3 DB "LL$"
```

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Q1:- X dB 2, y dB -8, z dB 9 .... write a program that finds the maximum number and stores it in memory location "maximum"

→ Solution:-

```
X dB 2
y dB -8
z dB 9
maximum dB ?
mov maximum, x
cmp maximum, y
JBE L1
cmp maximum, z
JBE L2
JMP L3
L1: mov maximum, y
    cmp maximum, z
JBE L4
JMP L3
L4:
L2: mov maximum, z
L3: HLT
```

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SOL. EXAM

PERIOD

NOTE

Q2:- Array dB 1, 2, -4, 9, -8 write a program that converts negative numbers to positive numbers in array

→ Solution:-

```
mov si,0
mov cx,5
L3: test array[si], 80H
    JNZ L1
    Inc si
    JMP L2
    L1: Neg array[si]
        Inc si
    L2:
Loop L3
HLT
array dB 1, 2, -4, 9, -8
```

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Q3:- Using logical gates write a program to find 2*N where N dB 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

→ Solution:-

```
mov si,0
L1 : SHL N[si]
Inc si
JMP L1
HLT
N dB 1,2,3,4,5,6,7,8,9,10
```

Q4:- Write an equivalant program to do the following:-

a) mov cx, bx
mov bx, ax
mov ax, cx

→ Solution:-

```
XChng ax, bx
```

b) Loopne

→ Solution:-

Don't miss the change..
Be part of it !!

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PERIOD

NOTE

mov cx, 10.

L2: .....

Dec cx

cmp cx, 0

JNZ L1

JMP L3

L1: JNC L2

L3: HLT

c) neg ah

→ Solution

Not ah

Inc ah.

Q5:- write a program that converts capital letters to small letters and vice versa in:- array dB 'A', 'b', 'C', 'D', 'e', 'F'

→ Solution:-

mov si, 0

L3: test array[si], 20H

JNZ L1

oR Array[si], 20 H

Don't miss the change..
Be part of it !!

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PERIOD

NOTE

JMP L2

L1:- And array[si], 0DF

L2: Inc si

Loop L3

HLT

Array DB 'A', 'b', 'C', 'D', 'e', 'f'

Q6:- X = 25 , y = 3 al = x/y ah = x % y write this program without using DIV or IDIV

→ Solution:-

```
mov CL,1
L3: mov AL,X
mul CL
cmp AL,y
JBE L1
Dec CL
mov AL,X
mul CL
JMP L2
L1: Inc CL
```

Don't miss the change..
Be part of it !!

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JMP L3

L2: mov AL, CL

sub y, AL

mov Ah, y

HLT

X dB 25

y dB 3

Q7: convert the Following C code to assembly code, where X and y are bytes (X=4, y=6). IF (X>=5) X = Y/3 else X = 2 - Y

→ solution

Don't miss the change..
Be part of it !!

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